



Computing Infrastructures
February 4, 2026

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

Exam Duration: 1hour and 30min

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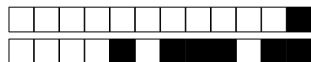
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Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.



True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 RAID 10 provides better random write performance than RAID 6.

☐ A True

☐ B False

Question 2 Edge computing always requires central data centers to process data.

☐ A False

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Question 3 Fragmentation is not a concern on HDDs since all data blocks are guaranteed to be sequential.

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Question 4 Hot/cold aisle containment reduces the efficiency of the cooling method for data-centers.

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Question 5 Monolithic hypervisors are more lightweight than microkernel-based hypervisors.

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Question 8 The power consumption of an average datacenter is in the order of a few KiloWatts.

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Question 9 Virtualization can improve disaster recovery by enabling quick and easy backup and restoration of virtual machines.

☐ A False

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Question 10 Neural network training benefits significantly from accelerators that can handle large-scale matrix operations.

☐ A True

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Exercises

Correct answer: +2, No answer: 0.

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Question 11

Consider an HDD with a data transfer time of 0.4ms for a 4 KB sector, a rotation speed of 12000 RPM, a mean seek time of 6 ms, and a negligible overhead of the controller. What minimum locality is required to achieve a mean I/O service time of 30 ms to transfer a 100KB file?

Question 12

A critical component used to implement an IT service is characterized by an MTTF = 640 hours and an MTTR = 8 hours. The availability of the service directly depends on the availability of this component, which represents a fundamental building block of the service architecture. The service can be implemented using two alternative configurations:

- Configuration A (single): the service depends on a single component.
- Configuration B (active redundancy): the service is implemented using two identical components and it is available if at least one of the two components is operational.

Given that service unavailability costs €800 per hour, estimate the expected annual cost savings (over a period of 365 days) in terms of downtime obtained by adopting Configuration B instead of Configuration A.



Question 13

We have to design a RAID 5 storage architecture composed of an array of 10 disks. Knowing that each disk has a MTTF equal to 600 days and that we would like to have a MTTF for the storage infrastructure ($MTTF_{RAID0+1}$) higher than 12 years, what is the maximum MTTR that we have to consider to satisfy the requirement? Consider all the disks with the same characteristics and a single mirror case for the RAID 1 part.

Question 14

A batch processing system consists of two components: a Processing Node (PN) and a Storage Node (SN). The system can be modeled as an open system. During a monitoring campaign lasting 3 hours, the following observations were collected:

- The average observed system throughput was 3 jobs per minute.
- The Processing Node showed an average utilization of 70%.
- The Storage Node showed an average utilization of 40%.
- Each job performs 5 accesses to the Processing Node and 1 access to the Storage Node.

Compute the average service time per visit S_{PN} and S_{SN} .



Question 15

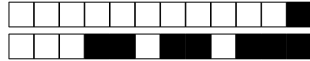
Determine the maximum throughput and the minimum response time the system can achieve.

Question 16

You are asked to compare two alternative configurations for an application cluster that can be modeled using a closed queuing model. On average, there are $N = 12$ users in the system, with a mean think time of $Z = 15$ s. Two alternative system configurations are considered:

- Configuration A: 8 CPU-based servers, each with a service demand of $D_{CPU} = 1$ s.
- Configuration B: 2 GPU-based servers, each four times faster than a single server in Configuration A, with a service demand of $D_{GPU} = 0.25$ s.

For both configurations, compute the value of N^* that identifies the transition point between the light-load and heavy-load regimes for the upper bound on throughput (the knee of the bounds).



+1/6/55+

Open Questions

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Question 17

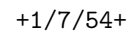
⇒ Are SSDs faster and more reliable than HDDs? Please provide a technical answer detailing the strong and weak points.

Question 18

⇒ Describe the concepts of under- and over-provisioning and how they can be mitigated by Cloud Computing.

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Answer Sheets (Page 1)

Student ID (Codice Persona):

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[illegible]





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Answer Sheets (Page 3)

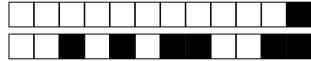
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True/False Questions

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Exercises

- Question 11 :
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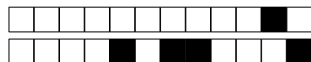
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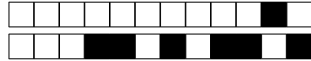
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+2/6/45+

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Question 10 : ☐A ☐B

Exercises

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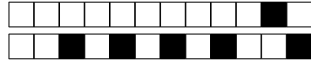
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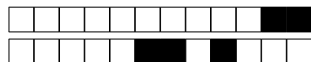
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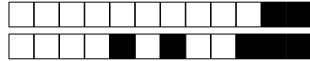
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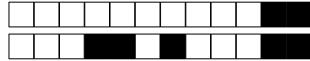
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+3/6/35+

Open Questions

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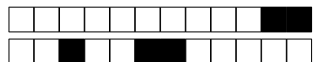
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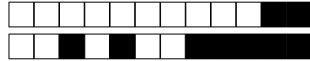
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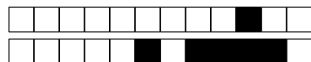
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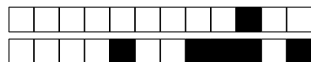
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True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 Neural network training benefits significantly from accelerators that can handle large-scale matrix operations.

☐ A True

☐ B False

Question 2 Open loop (free) cooling refers to the use of cold outside air to either help produce chilled water or directly cool servers.

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Question 4 Edge computing always requires central data centers to process data.

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Question 9 RAID 10 provides better random write performance than RAID 6.

☐ A True

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Question 10 Monolithic hypervisors are more lightweight than microkernel-based hypervisors.

☐ A False

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Exercises

Correct answer: +2, No answer: 0.

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Consider an HDD with a data transfer time of 0.5ms for a 4 KB sector, a rotation speed of 10000 RPM, a mean seek time of 6 ms, and a negligible overhead of the controller. What minimum locality is required to achieve a mean I/O service time of 40 ms to transfer a 100KB file?

Question 12

A critical component used to implement an IT service is characterized by an MTTF = 600 hours and an MTTR = 6 hours. The availability of the service directly depends on the availability of this component, which represents a fundamental building block of the service architecture. The service can be implemented using two alternative configurations:

- Configuration A (single): the service depends on a single component.
- Configuration B (active redundancy): the service is implemented using two identical components and it is available if at least one of the two components is operational.

Given that service unavailability costs €800 per hour, estimate the expected annual cost savings (over a period of 365 days) in terms of downtime obtained by adopting Configuration B instead of Configuration A.



Question 13

We have to design a RAID 5 storage architecture composed of an array of 10 disks. Knowing that each disk has a MTTF equal to 600 days and that we would like to have a MTTF for the storage infrastructure ($MTTF_{RAID0+1}$) higher than 12 years, what is the maximum MTTR that we have to consider to satisfy the requirement? Consider all the disks with the same characteristics and a single mirror case for the RAID 1 part.

Question 14

A batch processing system consists of two components: a Processing Node (PN) and a Storage Node (SN). The system can be modeled as an open system. During a monitoring campaign lasting 3 hours, the following observations were collected:

- The average observed system throughput was 3 jobs per minute.
- The Processing Node showed an average utilization of 70%.
- The Storage Node showed an average utilization of 40%.
- Each job performs 5 accesses to the Processing Node and 1 access to the Storage Node.

Compute the average service time per visit S_{PN} and S_{SN} .



Question 15

Determine the maximum throughput and the minimum response time the system can achieve.

Question 16

You are asked to compare two alternative configurations for an application cluster that can be modeled using a closed queuing model. On average, there are $N = 12$ users in the system, with a mean think time of $Z = 15$ s. Two alternative system configurations are considered:

- Configuration A: 8 CPU-based servers, each with a service demand of $D_{CPU} = 1$ s.
- Configuration B: 2 GPU-based servers, each four times faster than a single server in Configuration A, with a service demand of $D_{GPU} = 0.25$ s.

For both configurations, compute the value of N^* that identifies the transition point between the light-load and heavy-load regimes for the upper bound on throughput (the knee of the bounds).

+4/6/25+

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

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Question 17

⇒ Are SSDs faster and more reliable than HDDs? Please provide a technical answer detailing the strong and weak points.

Question 18

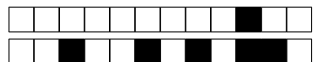
⇒ Describe the concepts of under- and over-provisioning and how they can be mitigated by Cloud Computing.

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●



Computing Infrastructures - February 4, 2026

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

Question 08 : ☐A ☐B

Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

Question 11 :

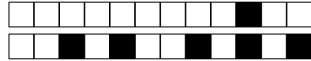
Question 12 :

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Question 14 :

Question 15 :

Question 16 :



+4/10/21+

Computing Infrastructures
February 4, 2026

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

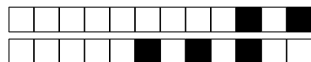
Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
(FIRST NAME IN CAPITAL LETTERS)

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Computing Infrastructures
February 4, 2026

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
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Exam Duration: 1hour and 30min

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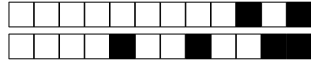
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**True false questions**

Correct answer: +1, No answer: 0, Wrong Answer -0.5

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Exercises

Correct answer: +2, No answer: 0.

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+5/4/17+

Question 13

We have to design a RAID 5 storage architecture composed of an array of 10 disks. Knowing that each disk has a MTTF equal to 600 days and that we would like to have a MTTF for the storage infrastructure ($MTTF_{RAID0+1}$) higher than 12 years, what is the maximum MTTR that we have to consider to satisfy the requirement? Consider all the disks with the same characteristics and a single mirror case for the RAID 1 part.

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Compute the average service time per visit S_{PN} and S_{SN} .



+5/5/16+

Question 15

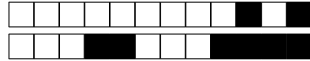
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For both configurations, compute the value of N^* that identifies the transition point between the light-load and heavy-load regimes for the upper bound on throughput (the knee of the bounds).



+5/6/15+

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

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Question 17

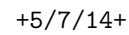
⇒ Are SSDs faster and more reliable than HDDs? Please provide a technical answer detailing the strong and weak points.

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Answer Sheets (Page 1)

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●



Computing Infrastructures - February 4, 2026

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

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Exercises

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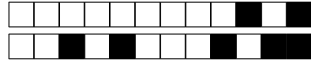
Question 12 :

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Question 15 :

Question 16 :



Computing Infrastructures
February 4, 2026

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

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First Name:
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+6/1/10+

Computing Infrastructures
February 4, 2026

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Exercises

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Consider an HDD with a data transfer time of 0.4ms for a 4 KB sector, a rotation speed of 12000 RPM, a mean seek time of 6 ms, and a negligible overhead of the controller. What minimum locality is required to achieve a mean I/O service time of 30 ms to transfer a 100KB file?

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+6/4/7+

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We have to design a RAID 0+1 storage architecture composed of an array of 10 disks. Knowing that each disk has a MTTF equal to 400 days and that we would like to have a MTTF for the storage infrastructure ($MTTF_{RAID0+1}$) higher than 8 years, what is the maximum MTTR that we have to consider to satisfy the requirement? Consider all the disks with the same characteristics and a single mirror case for the RAID 1 part.

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+6/5/6+

Question 15

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+6/6/5+

Open Questions

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Computing Infrastructures - February 4, 2026

Answer Sheets (Page 1)

First Name (CAPITAL LETTERS):

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Computing Infrastructures - February 4, 2026

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True/False Questions

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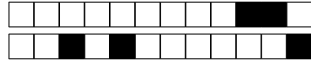
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+6/10/1+

Computing Infrastructures
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